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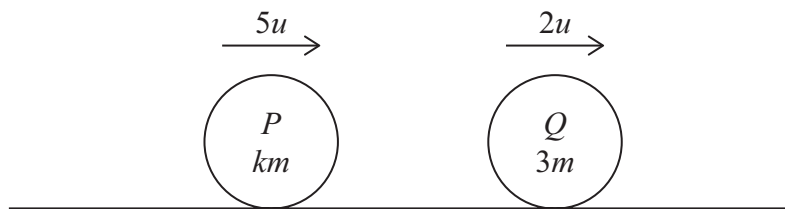


Figure 1

Figure 1 shows two particles, P and Q , moving in the same direction along the same straight line on a smooth horizontal surface.

Particle P has mass km and particle Q has mass $3m$

The particles collide directly.

Immediately before the collision, the speed of P is $5u$ and the speed of Q is $2u$

Immediately after the collision, the speed of P is $2u$ and its direction of motion is unchanged.

Immediately after the collision, the speed of Q is v

The impulse exerted on Q by P in the collision has magnitude $4.5mu$

(a) Find v in terms of u only.

(3)

(b) Find the value of k

(3)

2. [In this question $\mathbf{i}$ and $\mathbf{j}$ are horizontal unit vectors directed due east and due north respectively. Position vectors are given relative to a fixed origin.]

Two particles, A and B , are moving on a smooth horizontal surface.
Each particle is moving with constant velocity.

At time t seconds, the position vector of A is given by $\mathbf{r}$ metres.

- When $t = 2$, $\mathbf{r} = (-5\mathbf{i} + 16\mathbf{j})$
- When $t = 5$, $\mathbf{r} = (10\mathbf{i} + 4\mathbf{j})$

(a) Find, in terms of $\mathbf{i}$ and $\mathbf{j}$, the velocity of A .

(2)

(b) Find an expression for $\mathbf{r}$ at time t seconds.

Give your answer in the form $p\mathbf{i} + q\mathbf{j}$, where p and q are functions of t

(2)

At time t seconds, the position vector of B is given by $\mathbf{s}$ metres where

$$\mathbf{s} = -5\mathbf{i} + 7\mathbf{j} + t(2\mathbf{i} - 3\mathbf{j})$$

(c) Find, to the nearest degree, the bearing of B from A when $t = 5$

(3)

3. A particle is projected vertically upwards with speed $U \text{ m s}^{-1}$ from a point A .

The point A is 12 m vertically above the point B .

Point B is on horizontal ground.

The particle moves freely under gravity until it hits the ground at B .

The time taken for the particle to travel from A to B is 4 seconds.

(a) Find the value of U . (3)

(b) Find the speed of the particle as it hits the ground at B . (3)

(c) Sketch a speed-time graph for the motion of the particle from the instant it leaves A to the instant it reaches B . (No further calculations are required.) (2)

4.

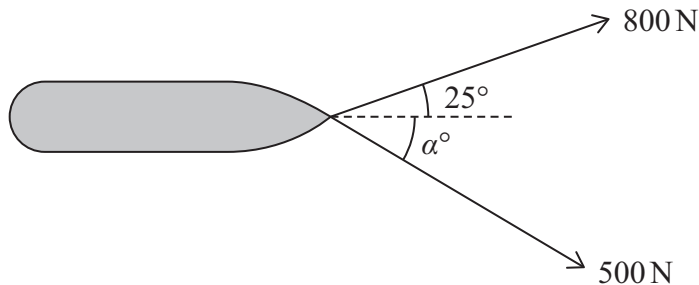


Figure 2

Two ropes are attached to a point on the front of a barge. The barge is being pulled horizontally in a straight line along the centre of a long straight canal.

One rope makes an angle of 25° with the direction of motion of the barge and has a tension of 800 N.

The other rope makes an angle of α° with the direction of motion of the barge and has a tension of 500 N, as shown in Figure 2.

Both ropes are horizontal.

(a) Find the value of α

(3)

The mass of the barge is 15 tonnes and the resistance to the motion of the barge is a constant force of magnitude 750 N.

(b) Find the acceleration of the barge.

(4)

5.

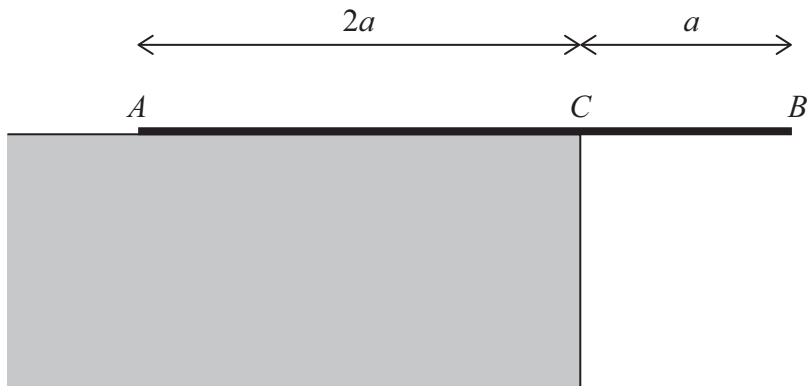


Figure 3

A non-uniform rod AB of length $3a$ rests in equilibrium on a horizontal ledge and overhangs the edge of the ledge at C .

The point C is such that $AC = 2a$ and $CB = a$, as shown in Figure 3.

The rod has weight W .

The distance of the centre of mass of the rod from A is d .

The rod is perpendicular to the edge of the ledge.

When a force of magnitude P , **acting vertically upwards**, is applied to the rod at B , the rod is on the point of tilting about A .

When the force applied at B is replaced by a force of magnitude $1.25P$, **acting vertically downwards** at B , the rod is on the point of tilting about C .

Find d in terms of a .

6.

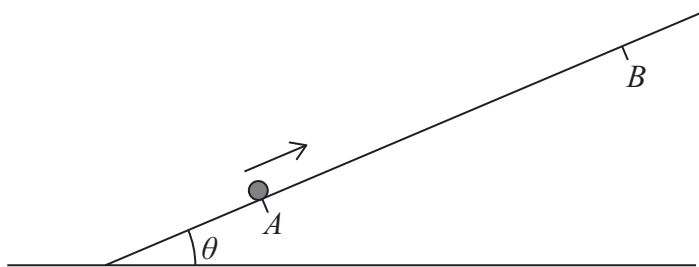


Figure 4

Figure 4 shows a rough plane inclined at an angle θ to the horizontal, where $\tan \theta = \frac{3}{4}$

The points A and B lie on a line of greatest slope of the plane, with B above A .

A package P of mass m is projected up the plane from A towards B .

The coefficient of friction between the plane and the package is $\frac{1}{4}$

The package P is modelled as a particle.

(a) Show that the **deceleration** of P , as it moves from A to B , is $\frac{4}{5}g$ (6)

The package P comes to rest at B .

Given that P is projected from A with speed $U \text{ m s}^{-1}$ and that $AB = 1.5 \text{ m}$,

(b) find the value of U . (2)

On reaching B , P is held at rest there by a force of magnitude X newtons acting up the plane in the direction AB .

Given that the mass of P is 2 kg ,

(c) find the smallest possible value of X . (3)

7. [In this question $\mathbf{i}$ and $\mathbf{j}$ are horizontal unit vectors.]

Two boats, A and B , are each moving with constant acceleration.

At time t hours after noon, boat A has velocity $\mathbf{v}_A \text{ km h}^{-1}$, where

$$\mathbf{v}_A = 2\mathbf{i} + 3\mathbf{j} + (\mathbf{i} - 4\mathbf{j})t$$

(a) Find the magnitude of the acceleration of A .

(2)

When $t = 2$, the velocity of B is $(4\mathbf{i} + \mathbf{j}) \text{ km h}^{-1}$

When $t = 5$, the velocity of B is $(\mathbf{i} - 5\mathbf{j}) \text{ km h}^{-1}$

(b) Find the acceleration of B , giving your answer in terms of $\mathbf{i}$ and $\mathbf{j}$.

(2)

(c) Find the velocity of B at time $t = 0$, giving your answer in terms of $\mathbf{i}$ and $\mathbf{j}$.

(2)

At time T_1 hours after noon, both boats are moving with the same speed.

(d) Find the exact value of T_1

(4)

At time T_2 hours after noon, both boats are moving in the same direction.

(e) Show that $3T_2^2 + pT_2 + q = 0$, where p and q are integers to be found.

(3)

8.

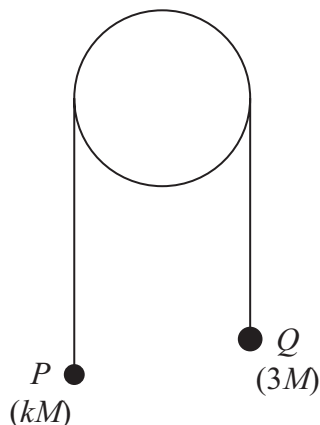


Figure 5

Two small balls, P and Q , have masses kM and $3M$ respectively, where $k < 3$

The balls are attached to the ends of a light inextensible string that passes over a fixed light smooth pulley.

The system is held at rest with the string taut and the hanging parts of the string vertical, as shown in Figure 5.

The system is released from rest and, in the subsequent motion, P moves with an acceleration of magnitude $\frac{1}{5}g$

The balls are modelled as particles.

(a) Write down an equation of motion for P .

(2)

(b) Find the value of k .

(3)

Given that $M = 0.5 \text{ kg}$,

(c) find the magnitude of the force exerted on the pulley by the string while Q is moving downwards.

(3)

At the instant when the system is released, P is more than 2.5 m from the pulley and Q is 2.5 m above horizontal ground.

After hitting the ground, Q rebounds with a speed of 0.4 m s^{-1}

(d) Find the magnitude of the impulse received by Q when it hits the ground.

(5)

In the subsequent motion, P does not hit the pulley.

(e) Find the total time from when the balls are released until P first comes to rest.

(4)